Step 1: Flawed Original Proof Excerpt

Flawed Original Proof

Hence, we can apply Glick’s extension to Scheffe's Lemma (e.g. Devroye and Wagner(1979)) to obtain

$$\int_{\mathbb{R}^d} |\psi_{n+1}^N(x) - \psi_{n+1}(x)| dx \overset{a.s.}{\longrightarrow} 0$$

Typographical artifact:
Ambient dimension

from which we can conclude […] almost surely as $N \to \infty$.

Convergence of the MISE is a consequence of Proposition 8, (A0) and the Dominated Convergence Theorem

$$\mathbb{E}\left[\int_{\mathbb{X}} \left|f_{n+1}^N(x) - f_{n+1}(x)\right|^2 dx\right]$$
$$\leq \lambda(\mathbb{X})\mathbb{E}\left[\left\|f_{n+1}^N - f_{n+1}\right\|_\infty^2\right] dx \to 0$$

Logical gap +
Extraneous dx

as $N \to \infty$, where $\lambda(\mathbb{X}) < \infty$.

Step 2: Localized Patching Execution

Localized patches

- **Patch 1** (Typographical Correction): Replace $\mathbb{R}^d$ with $\mathbb{R}^{dx}$.
- **Patch 2** (Syntax Repair & Gap Filling): Remove the extraneous dx and insert the missing intermediate supremum bound.

$$\mathbb{E}\left[\int_{\mathbb{X}} \left|f_{n+1}^N(x) - f_{n+1}(x)\right|^2 dx\right]$$
$$\leq \mathbb{E}\left[\int_{\mathbb{X}} \left\|f_{n+1}^N - f_{n+1}\right\|_\infty^2 dx\right]$$

Zero-Deletion Policy: *Only localized additive corrections; surrounding proof text remains fixed.*

Step 3: Synthesized Refined QA

Question

Assume (A0), (A1), and (A2) hold, $s_N \to 0$, and $\mathbb{X} \Subset \mathbb{R}^{dx}$ with $\lambda(\mathbb{X}) < \infty$. Prove that f_{n+1}^N converges almost surely to f_{n+1} in L_1, and that

$$\lim_{N\to\infty} \mathbb{E}\left[\int_{\mathbb{X}} \left|f_{n+1}^N(x) - f_{n+1}(x)\right|^2 dx\right] = 0$$

Reference Proof

[Initial proof steps omitted for brevity…]

Hence, we can apply Glick’s extension to Scheffe's Lemma … to obtain

$$\int_{\mathbb{R}^d} |\psi_{n+1}^N(x) - \psi_{n+1}(x)| dx \overset{a.s.}{\longrightarrow} 0$$

Convergence of the MISE is a consequence of Proposition 8, (A0) and the Dominated Convergence Theorem

$$\mathbb{E}\left[\int_{\mathbb{X}} \left|f_{n+1}^N(x) - f_{n+1}(x)\right|^2 dx\right] \leq \mathbb{E}\left[\int_{\mathbb{X}} \left\|f_{n+1}^N - f_{n+1}\right\|_\infty^2 dx\right]$$
$$= \lambda(\mathbb{X})\mathbb{E}\left[\left\|f_{n+1}^N - f_{n+1}\right\|_\infty^2\right] \to 0$$

as $N \to \infty$, where $\lambda(\mathbb{X}) < \infty$.